Guidance note on Climate Impact Potential for FAO-GCF projects

Version 1 – Updated on 8 March 2024

Summary

This note is intended to provide guidance on how to demonstrate climate impact potential of Green Climate Fund (GCF) projects at GCF Concept Note (CN) stage and the Funding Proposal (FP) stage.

The GCF funds **only** climate change adaptation and mitigation projects so the project design needs to be constructed around a rationale for investments that modulate climatic and non-climatic contributing factors to climate risks.

It is crucial to understand the agroclimatic context, determine climate risks and impacts and identify interventions that address climate change challenges for targeted areas/sectors by using the best available information. The proposal needs to clearly present linkages between identified climate hazards and impacts, and propose climate change responsive actions that address identified climate risks with highest societal, economic, and environmental benefits.

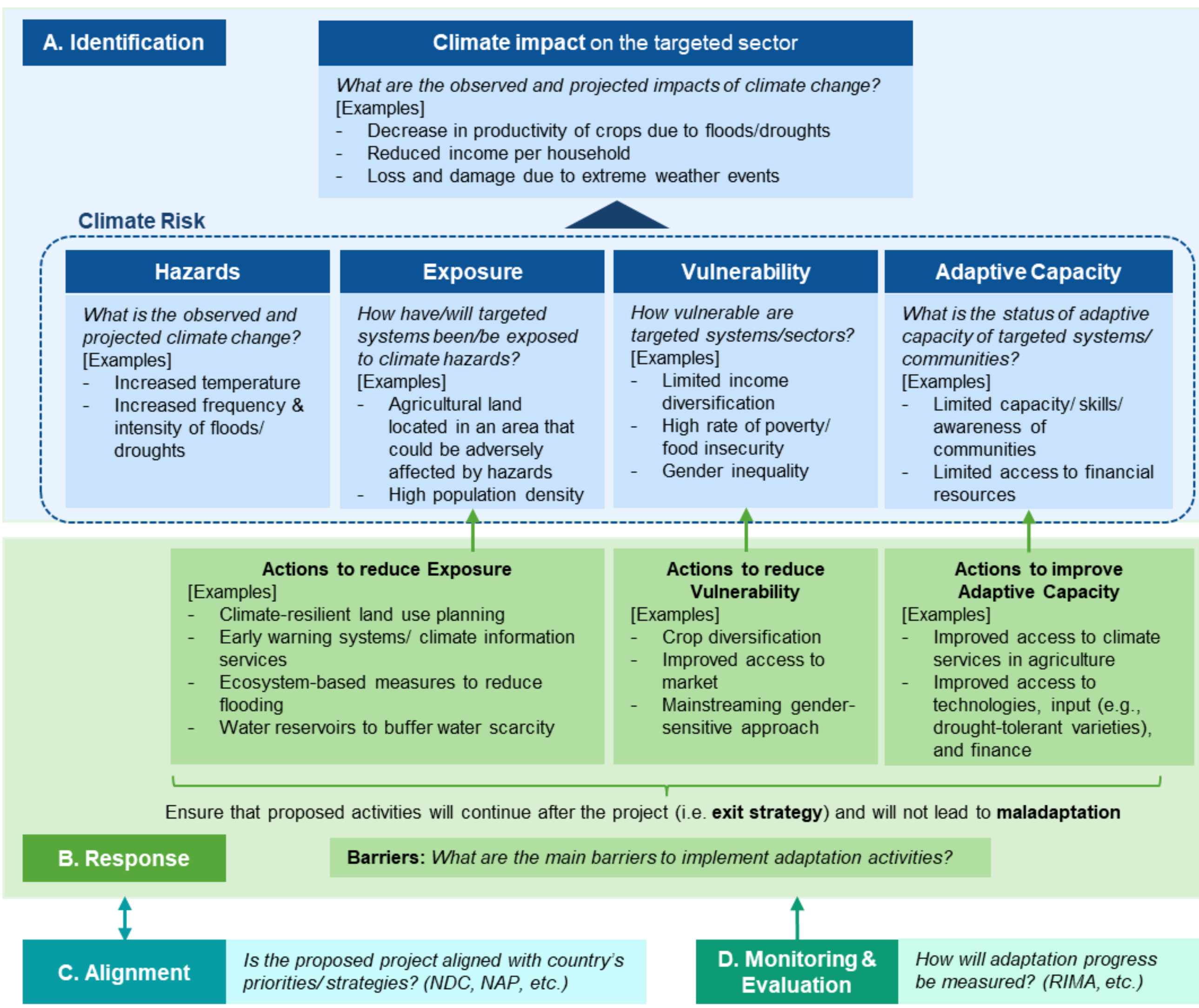
The selection of project areas should be justified based on emerging findings of the climate risk assessments.

In this guidance note, you will find:

- **What is climate impact potential?** (see section 1.2): The climate impact potential is assessed as the potential contribution of the project's interventions to adaptation and/or mitigation results.
- **How does the GCF assess it?** (section 2): The review of a CN by the GCF focuses firstly on climate impact to determine whether the project is designed to contribute primarily to low-emission and climate resilient development pathways, and whether it can demonstrate, with confidence, that it will provide measurable mitigation and/or adaptation results.
- **What is the GCF guidance on the climate impact potential?** (section 3 and Annex 2): The GCF has established high-level principles for demonstrating the impact potential of GCF-supported activities (GCF/B.33/05 entitled "Steps to enhance the climate rationale of GCF-supported activities").
- **What are the tools/information sources available?** (see Annex 1)
- **Which questions need to be answered for mitigation?** (section 3.23.2 Mitigation):
    - *What is the emissions profile?*
    - *What are the regulatory frameworks in place to reduce emissions potential?*
    - *What are the mitigation needs?*
    - *What are the barriers to implement mitigation activities?*
    - *What kind of activities would be implemented to reduce emissions/sequester carbon?*
    - *What is the emission reduction potential of the project activities using appropriate methodologies for GHG calculation?*

- **Which questions need to be answered for adaptation?** (section 3.1):
    - *What are the main climatic changes that have occurred over time in the past? How is the climate going to change in the future?*
    - *What are the observed and projected impacts of climatic changes?*
    - *How have/will targeted systems been/be exposed to climate hazards?*
    - *How vulnerable are targeted systems/communities to the climatic changes? What is the status of their adaptive capacity to cope with climate challenge?*
    - *Where is the most vulnerable area in the country? Why has the project target area been selected from the climate change perspective?*
    - *What are the adaptation components this project seeks to implement to address identified climate risks and barriers that prevent adaptation activities?*
    - *Is the proposed project aligned with country's priorities/ strategies? What current climate-related coping strategies/ interventions are in place and how will the proposed GCF project complement and/or address major gaps?*
    - *How will adaptation progress be measured? What are the indicators to be monitored?*

Approach proposed in this guidance note for adaptation projects:



Contents

1. Introduction

1.1. Objectives of this guidance note

This note provides guidance on how to articulate climate change challenges and proposed interventions with sufficient information/data to receive positive feedback from the GCF against its investment criteria and appraisal areas.

The target readers of this note are mainly project formulators and FAO Country Offices, but also reviewers (including Climate Finance Officers) and Lead Technical Officers (LTO) to understand GCF requirements on the climate impact potential and key steps to demonstrate the impact potential of proposed projects.

1.2. What is the climate impact potential?

The climate impact potential[1] is assessed as the potential contribution of the project's interventions to the achievement of the GCF's strategic objectives with respect to adaptation and mitigation.

The GCF only funds climate change adaptation and mitigation projects. Therefore, the project design must be constructed around a rationale of interventions responding to climate change adaptation and mitigation needs by modulating the observed and expected risks, based on the climate impact potential assessment that provides a better understanding of climate change impacts.

2. Assessment of the climate impact potential by the GCF

2.1. How does the GCF assess the climate impact potential?

The **climate impact** is one of the ten appraisal areas the GCF looks at during its review process and is directly related to one of GCF's investment criteria "impact potential''. In this line, the GCF Appraisal Guidance (published in July 2022) clearly indicates that the review of a CN by the GCF focuses firstly on climate impact to determine whether the project is designed to contribute primarily to low-emission and climate resilient development pathways, and whether it can demonstrate, with confidence, that it will provide measurable mitigation and/or adaptation results. The first review also covers the project's Theory of Change and examines whether the proposed interventions modulate the identified climate risks by minimizing the exposure and vulnerability of targeted agricultural systems to climate hazards (see Box 1 for core concepts of climate risk).

Adaptation: appraisal for this area focuses on climate risks and vulnerability assessments and its linkages with the proposed interventions. Sufficient information must be provided to generate confidence that the proposed intervention will reduce the exposure and address the vulnerabilities to climate hazards, besides contributing to the wellbeing of the affected populations.

The GCF has preliminarily identified a checklist of guidance questions;

- *How does the proposal show convincingly how climate change has led, or will lead, to the specific risk/impact that the proposed activity addresses?*
- *How does the proposal describe the observed and projected future changes of the climate hazard specific to the climate risk/impact identified?*
- *How does the proposal describe the exposure and vulnerability of people, systems or ecosystems to the specific climate hazard?*

[1] The terminology ''climate rationale'' has been often used in GCF-related discussions. However, given that ''climate rationale'' is not new project selection or eligibility criteria for GCF funding, a working definition of ''climate rationale'' has been made in GCF/B.33/05 as the overarching narrative of a proposal which ensures that it addresses climate change and thus contributes to the criteria of the GCF Investment Framework. Instead of ''climate rationale'', the GCF is currently using ''climate change basis/context''.

- *How does the proposal consider the non-climate related drivers that may interact with climate variability and change?*
- *How were activities selected as the best to tackle the climate hazard, exposure and vulnerability identified?*
- *How does the proposal show how the activity will enhance adaptive capacity, strengthen resilience and/or reduce vulnerability to climate change?*
- *How has the proposal determined the expected number of beneficiaries, who are those beneficiaries and precisely how will they benefit?*
- *How does the proposal provide information and analysis to show that the proposed activity will not lead to maladaptation? (see IPCC AR6 WGII SPM.C4 and references therein)*
- *How will the proposed activity strengthen institutional capacity to improve climate-resilient practices and drive policies that reduce climate risk?*

Mitigation: appraisal for this area focuses on an estimation of baseline emissions and the projections for the reduction or avoidance of net emissions as a result of the proposed interventions.

Box 1. Core concepts of climate risk[2]

Climate risk results from the interaction of climatological, hydrological, and meteorological **hazards**, with the geographical **exposure** of human and natural systems, together with **vulnerability** to climate hazards. The interactions between hazards, exposure, and vulnerability are also modulated by **the adaptive capacity**, which reduces the overall level of risk. Climate risk can be partly or totally offset both in the short- and long-term by implementing climate adaptation and mitigation strategies (IPCC, 2022). The risk framework is based on the Intergovernmental Panel on Climate Change's (IPCC) most up-to-date definition of risk from the Technical Summary of the Special Report on the Ocean and Cryosphere in a Changing Climate (IPCC, 2019). The understanding of risk is also based on the IPCC's risk conceptual framework used in the 5th Assessment Report (AR5) (IPCC, 2014), as well as in the 6th Assessment Report (AR6) (IPCC, 2022).

Hazard refers to the potential occurrence of a natural or human-induced physical event or trend that may cause loss of life, injury or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems and environmental resources.

Human and natural exposure to climate hazards is determined by the presence of people; livelihoods; species or ecosystems; environmental functions, services, and resources; infrastructure; or economic, social or cultural assets in places and settings that could be adversely affected by a hazard. The exposure component normally assesses the geographical presence of agro-ecological systems, such as crops, shrubs, forests, grassland, mangroves, fishery biomass, protected areas, lowland areas, and population density in the selected area.

Vulnerability refers to the propensity or predisposition to be adversely affected. Vulnerability encompasses a variety of concepts and elements, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt. Vulnerability assesses development, poverty, and gender equality indices, conflicts, migration, food insecurity, economic, and health indicators.

Adaptive capacity lies in actors (systems, institutions, humans and other organisms)' ability to adjust to potential damage, to take advantage of opportunities or to respond to consequences. It largely depends on the support provided by public and private institutions through research and investment in technological developments and social, agricultural and disaster risk insurance programmes, as well as on communities' means of adaptation. The adaptive capacity component assesses the access to climate

[2] IPCC AR6 WGII Annex II Glossary

information, electricity and internet, infrastructure development, as well as national institutional support through policy and financial mechanisms for climate-resilient agriculture.

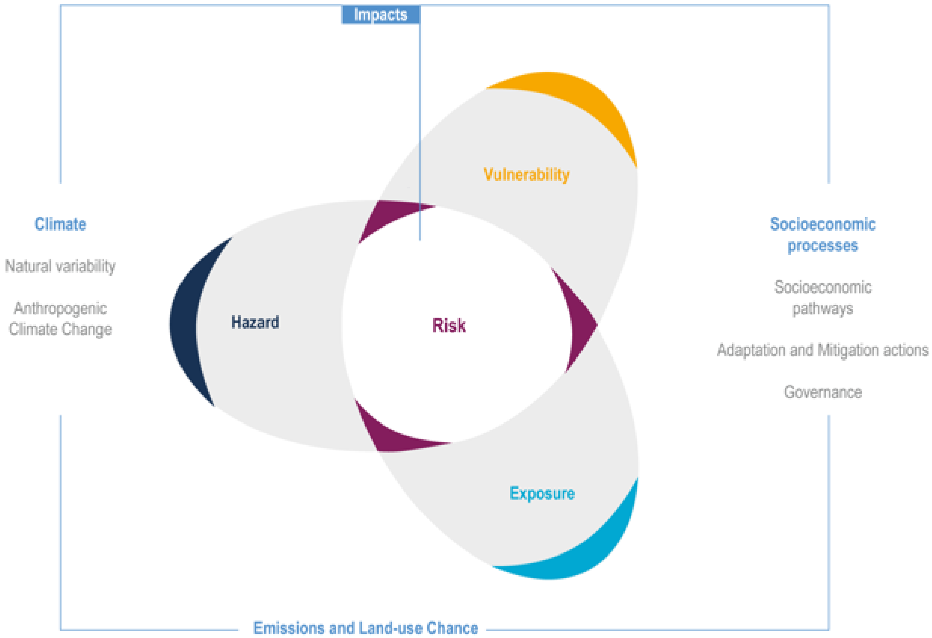


Figure 1. Concept of risk (IPCC)[3]

3. Key steps to demonstrate the climate impact potential

The GCF has established high-level principles for demonstrating the impact potential of GCF-supported activities (GCF/B.33/05 entitled "Steps to enhance the climate rationale of GCF-supported activities", also see Annex 2 of this note). In alignment with those high-level principles, the following guiding questions help to understand the climate context and linkages of climate change challenges with the proposed interventions. Also refer to the presentation (July 2023) by independent Technical Advisory Panel (iTAP) on impact potential for adaptation.

3.1 Adaptation

The following section provides the key steps to demonstrate the climate impact potential for adaptation projects in alignment with four high-level principles (Identification, Response, Alignment, Monitoring and evaluation) described in GCF/B.33/05.

A. Identification

The first step to demonstrate climate impact potential for adaptation project is to identify the systems at higher risk, within a country or countries, and the climate change hazard affecting them or expected to affect them in the future.

In the case of small countries (or projects that cover an entire country), it would be enough to present results of climate analyses at the national level. For large countries, it is recommended to look at sub-national results. For SIDS, it might be necessary to use regional information or adapt global information to be able to capture the land/water interactions.

[3] IPCC, AR6, 2022 (https://www.ipcc.ch/report/ar6/wg2/figures/chapter-1 , fig 1.5a)

A-1. Climate hazards: historical observations and future projections

First of all, the proposal needs to evaluate how climate change has evolved in the past (historical observations) and how the climate will change in the future (future projections).

a. *What are the main climatic changes that have occurred over time in the past (long-term changes in precipitation, temperature, sea level, ocean chemistry, etc.) as well as extreme weather events over the same time period (e.g. droughts, floods, storms and landslides)? How is the climate going to change in the future compared to the climate baseline (average climate over the last decades) and past events?*

Use data on historical observations (e.g. ground observations, gridded datasets, reanalysis) and projections (e.g. simulations) for outlining long-term changes and associated changes in extreme weather events. Note the difference between climate change (trends over at least two decades, ideally three decades) and climate variability (at various time scales from day, seasonal, inter-annual, to inter-decadal).

For future climate projections, the outputs from CMIP5 or CMIP6 global model inter-comparison initiatives and downscaled projections from the CORDEX initiative, and similar climate projection datasets specifically prepared for the country are recommended. At least, two RCPs (Representative Concentration Pathways) should be used, typically RCP 2.6 or 4.5 (low or moderate emission scenario) and RCP 8.5 (higher emission scenario)[4]. Make sure you add references to the RCPs, climate models, and downscaling methods that are used. Different climate models show different sensitivities to greenhouse gas emission changes and, therefore, highlighting the differences amongst them is also recommended. Wherever possible, consult projection results from two or more climate models to understand the uncertainty ranges in climate variables change. Cross-check the climate projections with the WMO-GCF climate information portal.

To note, projected long-term changes will always have uncertainty as they depend on future emission scenarios, how all other variables will change and modulate climate in different socio-economic pathways, scientific understanding of how climate will respond to these changes, and parameterization of climatic models to simulate these changes. This should be reflected in the narrative used when drafting the CN.

Where to find data/information?

- For information on observed climatic changes at the national and sub-national level, project design teams should reach out to national institutions – for example via the FAO country office or the NDA - as local climatic data is often available at the National Meteorological and Hydrological Services (NMHS).
- NMHS may also have downscaled future climate projections.
- Please also refer to the National Communication to the UNFCCC, and other national climate change reports where these historical and future climate data are presented.
- A wide range of tools and information platforms exist to assist in the retrieval and analysis of climate model information (see Annex 1). FAO's CAVA[5] (Climate and Agriculture Risk and Visualization Assessment) allows users with a wide range of expertise to access regional and global gridded climate data (future projections and observations for most part of the world) with the possibility of generating automatic report that satisfy many of the impact potential assessment requirements.
- The FAO Risk Team within the Office of Climate Change, Biodiversity and Environment (OCB) has the expertise to conduct the climate adaptation impact potential assessments that includes the analysis of historical climate observations and projections, besides making the linkages between climate hazards and impacts on targeted agricultural systems, most effective adaptation interventions with highest societal, economic, and environmental benefits. Please contact Jorge.AlvarBeltran@fao.org copying Hideki.Kanamaru@fao.org.

[4] In case RCP 4.5 is not available for certain datasets (such as CORDEX CORE information), RCP 2.6 can be used instead of RCP 4.5 with justification.
[5] https://fao-cava.predictia.es/ and https://github.com/Risk-Team/CAVAanalytics/blob/main/README.md

A-2. Observed and projected climate change impacts on the targeted systems/sectors

Based on the assessment of climate change (A-1), the proposal should assess how climate change (e.g. increased temperature, increased intensity and frequency of droughts and floods, etc.) has affected and will affect the targeted systems/sectors (e.g. decreased productivity of crops due to drought leading to reduced income of smallholder farmers).

b. What are the observed and projected impacts of climatic changes? It is important to briefly outline the estimated climate change impacts (both past and future) on the targeted sector(s).

For example, characterization of climate change impacts may look like this for crops; The livelihoods of local communities in governorate X who depend on rice variety Y are very vulnerable to observed and projected changing climatic conditions. The historical rice yield data indicate a general trend of decline over the past S years, which has reduced income per household and resulted in the lower intake of calories across the population. Without introducing context-specific adaptation measures, the rice yield in governorate X is projected to further decline by 30% by 2030 (relative to year Z) under climate change, with severe implications for the local communities.

Where to find data/information?

- The project design team should search for up-to-date research publications/studies regarding the impacts of climate change on targeted sectors.
- Please look for agricultural statistics of the country, for example, changes in crop production, and area harvested (and resulting yield) over the last years. Please also consult national data on disaster loss and damages in agriculture. See Annex I for some of the global databases, which may complement national data.
- Other useful tools/information sources are listed in Annex 1.
- To note, information on projected climate change impacts on ecosystems, the agriculture sectors, post-harvest systems and livelihoods is not always easily available. The FAO-OCB Risk Team can provide direct support on climate impact assessments by using impact models output of ISIMIP and other sources. In addition, a crop impact assessment with GAEZ for the past can be done at concept note stage.
- At the FP stage, an analysis of future changes in crop productivity may be useful using PyAEZ, AquaCrop, or other tools, depending on the project design and the level of details required. Watch an introductory video of the University of Edinburgh on the importance of cross-sectoral climate impact modelling here.

A-3. Climate risk for targeted systems: exposure, vulnerability and adaptive capacity

As explained in Box 1, the level of climate risk is determined by dynamic interactions between climate-related hazards with the exposure and vulnerability of the affected human or ecological system to the hazards. For example: based on downscaled projections and modelling for country X (see A-2), area X is exposed to climate hazards such as higher temperatures, drier conditions, increased evapotranspiration, and late onset of rainy season. Climate projections indicate that droughts will become more frequent and intense. Rice variety Y, the main staple crop in area X, is exposed and vulnerable to these observed and projected climatic conditions. The natural resilience of the rice variety as well as the adaptive capacity of local farmers to cope with the changing climatic conditions is limited due to unavailability of drought resistant rice varieties (see A-5).

c. Exposure: How have/will targeted systems been/be exposed to climate hazards?

Describe how targeted systems (ecosystems, the agriculture sectors, livelihoods, etc.) in a location have been (or will be) exposed to different kinds of climate hazards identified in A-1 with quantitative data as much as possible. The level and characteristics of exposure can be analyzed from historical observed impacts, for example: the severe and prolonged drought of years TT affected XX million people, causing damages and losses worth USD YY billion in the amount of ZZ tons of crops in the area of WW ha, or number of animals lost, etc.

d. *Vulnerability and adaptive capacity: How vulnerable are ecosystems, the agriculture sectors (crops, livestock, forestry, fisheries, and aquaculture), post-harvest systems, marketing opportunities, livelihoods, food security and nutrition to the climatic changes described in A-1.? What is the status of adaptive capacity of targeted systems/ communities to cope with climate challenge?*

Vulnerability can be explained by the social and economic conditions of the targeted systems and population.

The adaptive capacity includes human capacity (UNDP Human Development Index, literacy rates, etc.), technical skills and knowledge, available financial resources, available and implemented adaptation options, etc.

Non-climatic barriers/drivers that can exacerbate climate risks as a vulnerability or adaptive capacity factor include:

A) Conflict: Conflict is one of the factors which hinder climate change adaptation activities. At the same time, climate threats could amplify conflict risks especially in fragile countries.
B) Institutional barrier: weak institutions are one of the main causes of poor implementation of climate action. In countries where the role of national institutions is weak, the implementation of such policies by sub-national governance level is limited.
C) Local and technical knowledge: extension workers and farmers' technical capacity to adapt to climate change can be inadequate in some contexts and, thus, hampering capacity to build climate resilience, including planning, programming, monitoring and resource mobilization.
D) Gender: in some contexts, female farmers rarely have ownership rights over the land and commonly renounce land rights to male family members, which increases their vulnerability.
E) Social and cultural barrier: community adaptation capacity is limited due to the low access to information, markets, services, and technology required for climate-resilient livelihoods, especially for women and other vulnerable groups.
F) Financial barrier: in some decentralized countries, the capacity of sub-national level institutions to mobilize climate finance and to roll out adaptation and mitigation efforts is reduced.

The GCF recognizes the importance of considering future socio-economic conditions to understand future exposure and vulnerability[6]. For example, future exposure to climate hazards could be different if the expected population in the area by 2050 is 10,000 people or 100,000 people.

Please note that the GCF will not invest in activities focused on dealing with only non-climate drivers. Therefore, those non-climate drivers should not be the main focus of the project rationale as a whole.

There are many different approaches/ methodologies to analysing and assessing vulnerability including FAO's Self-evaluation and Holistic Assessment of Climate Resilience of Farmers and Pastoralists tool (SHARP+), CGIARs toolkit in working paper No. 108, the GIZs and the Government of India's A Framework for Climate Change Vulnerability Assessments, and/or IISDs CRiSTAL, which is a project planning management tool that helps users to integrate risk reduction and adaptation into their community level work.

The assessment of the vulnerability and adaptive capacity of the target project area, agricultural systems, and value chains using existing tools and databases can be complemented by the delivery of climate-sensitive consultations with target stakeholders in the project area, aimed at assessing stakeholders' perception of climate and weather-related hazards and impacts on agrifood production and post-harvest systems, the capacities to implement climate resilient practices, as well as the key socio-economic challenges to adopt climate resilient measures and benefit from climate and weather services and products, which are important instruments to strengthen adaptive capacity, and an essential pillar within

[6] The iTAP's presentation in July 2023.

the range of measures available to reduce climate risks, support livelihoods, protect assets and improve food security and nutrition,

A-4. Target area

As the GCF puts specific focus on supporting countries, and their communities and people, who are most vulnerable to the adverse effects of climate change, the most vulnerable areas should be selected as the project target area.

e. Where is the most vulnerable area in the country? Why has the project target area been selected from the climate change perspective?

The project target area has to be selected with a strong justification from the climate change point of view, i.e. the most vulnerable area based on the analysis through *A-1 to A-3* above needs to be selected.

Box 2. How to justify the selection of the project area?

There are mainly two ways to justify the selection of the project area for GCF projects; 1) select areas based on a climate risk assessment of the entire country or 2) select the main agricultural areas that are the most important for the country and vulnerable to the adverse effects of climate change.

1) Select areas based on a climate risk assessment of the entire country

As per the latest IPCC risk framework (refer to Box 1), the project target area can be selected based on a climate risk assessment which evaluates hazards, exposure, vulnerability, and the existing adaptive capacity. The exercise for selecting the project area can be done, for example, by creating maps (Figure 2) or scoring for each region/province (Figure 3).

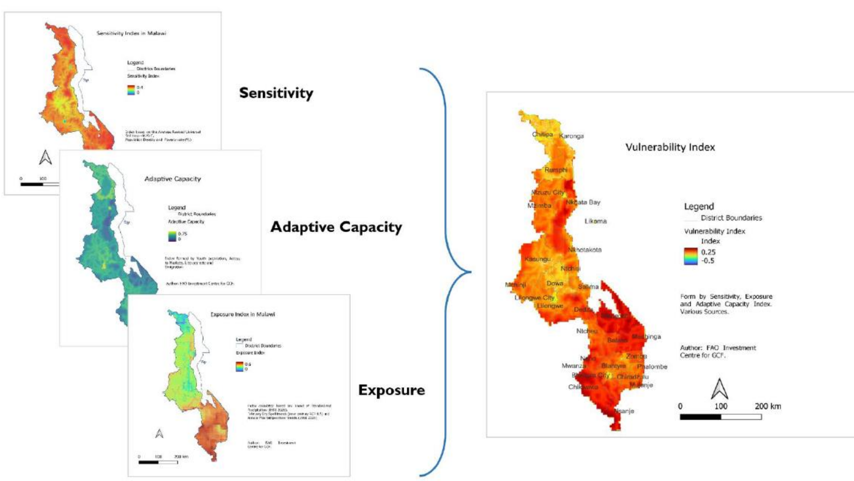


Figure 2. Identification of high vulnerability areas (an example from Malawi[7])

[7] Not yet approved by the GCF Board.

Geographic climate change impact regions	Average of Decrease Rainfall + Increased Temp	Average of Increase Cyclone + Precipitation	Average of Combined CC physical impact	Notes on application of significant selection criteria	Selection as project area
1. Cordillera mountains (complex combined effects)	0.41	0.57	0.49	Combined effects high	Yes
2. Luzon foodbasket (increased precipitation, intensified cyclones)	0.29	0.60	0.45	High cyclone and precipitation	Yes
3. Eastern Seaboard (intensified cyclones, increased precipitation)	0.37	0.47	0.42	High cyclone and combined effects	Yes
4. Central islands (combined effects)	0.39	0.23	0.31	While CC expected, less exposure than other regions	
5. Eastern Mindanao (decreased precipitation, increased temperatures)	0.63	0.17	0.40	Ongoing plans for other climate project at the time of GCF design	
6. Western Mindanao (decreased precipitation, increased temperatures)	0.64	0.04	0.34	Greatest expected decrease in precipitation and increased temperatures	Yes
Grand Total	0.44	0.33	0.39		

Figure 3. Relative severity of different types of climate change risks comparing all regions in the country (an example from the Philippines)

FAO Climate Risk Toolbox can also produce climate risk maps aggregated by the different components of risk (hazard, exposure,and vulnerability) with national and sub-national level data.

2) Select the main agricultural areas that are the most important for the country and vulnerable to the adverse effects of climate change

In the case of selecting the project area at the basin level which is a key area for the agriculture sector in the country with evidence (e.g. the total production in the area vs. other areas) and is vulnerable to climate change based on the analysis through A-1 to A-3, the detailed comparison of vulnerability with other areas (outside of the basin) may not be necessary (an example from the approved FAO-GCF projects: Benin). However, the criteria for selecting targeted districts/specific locations is required in terms of climate change vulnerability.

B. Response

B-1. Proposed interventions

f. *What are the adaptation components this project seeks to implement to address above outlined climate risks? And how will the proposed adaptation components contribute to increasing resilience to climate change and climate variability and their impacts on agriculture?*

For adaptation projects, response actions are planned to reduce adverse climate impacts by reducing exposure to phenomena that are exacerbated by climate change, reducing vulnerability of XX, and enhancing adaptive capacity, and consequently to increase resilience to climate change. Resilience focuses on the dynamic capacity of a system to recover from the impacts of climatic changes and adapt itself in the changing environment in the long term. Resilience refers not only to shocks (a change relative to an average), but also to the change of the average itself.

The proposed interventions must be clearly linked with risks analysed in section A and barriers need to be elaborated in B-2. Multiple adaptation interventions may be identified, and they may need to be prioritized using an option analysis through a set of objective criteria and/or cost-benefit analysis to justify the proposed interventions as the best possible adaptation measures.

In addition, GCF recognizes the role that weather and climate services and information products play as important instruments to strengthen adaptive capacity, and an essential pillar within the range of measures available to reduce climate risks, support livelihoods, protect assets and improve food security and nutrition.

B-2. Barriers

g. *Why has the adaptation to climate change in the project area not happened yet? What are the main barriers to implement adaptation activities and how will these be addressed by the project?*

Provide concrete examples for the main barriers (for example, farmers are not investing in the crop varieties resistant to drought because these are not available in the local markets, or the improved varieties need to be introduced together with technical and extension assistance on how to manage this new variety). Describe barriers that are social, gender, fiscal, regulatory, technological, financial, ecological, institutional, etc. as needed. All barriers mentioned in this section should be included in the Theory of Change diagram and be addressed by the project. It is therefore important to contain the analysis within the scope of what the project will be able to do.

B-3. Maladaptation risks

Adaptation barriers closely relate to maladaptation risks due to the interlinkages between resources, capacities and coordination which can have a significant impact on adaptation practices and their outcomes. Maladaptation, as a concept, generally refers to adaptation initiatives that cause additional vulnerabilities instead of diminishing them. Even though initiatives can foster adaptation in the short term, there is a chance that they may affect territories, sectors, and peoples' medium or long-term capacity to manage climate change impacts. The proposal should discuss that the proposed activities will not lead to maladaptation and how maladaptation risks can be mitigated.

Table 1. A sample summary of ''A. Identification'' and ''B. Response'': Climate risks and farming systems, and adaptation interventions (an example from the Philippines, FP201)[8]

Region	Major farming systems	Climate variable	Historical trend	Future projection (primary)	Secondary climate impact	Adaptation interventions (and mitigation, if any)	Adaptation benefits (and mitigation, if any)
North East Luzon	1) Rice lowland farming 2) Corn lowland farming	Rainfall	Slight increase in rainfall in DJF and MAM and slight decrease in JJA and SON between 1951-2010	Increase in precipitation in all seasons by >50% by 2050 under RCP4.5	Increase in flooding during rainy season and increase in drought during dry season	Flood-tolerant varieties, timing of farm practices, farm infrastructure and design, disaster insurance	More climate-resilient production (i.e. flood), and higher and stable income for farmers
		Temperature	Decrease in number of cold nights (1951-2010)	Increase in temperature by up to 1.5C by 2050 under RCP4.5	Increase in extreme temperature, drought during dry season	Heat- and drought-tolerant varieties, timing of farm practices (shorter growing seasons), water managements and conservation, including alternate wetting and drying method, irrigation, seasonal and short-term forecasts	More climate-resilient production (i.e. heat and drought), and higher and stable income for farmers. Efficient water use. Reduced methane emissions.
		Tropical cyclone	Increasing intensity, slightly	Increase in TC intensity	Increase in flooding and	Early warning systems, Strengthened	Production damage and loss are reduced

[8] This table shows only one region just to give an example.

			decreasing frequency		damage due to TC	infrastructure, disaster insurance, integrated farming practices	through anticipatory actions.
		Sea level rise	Sea level rise between 1993-2015 up to 4.5-5 mm per year (UK Met)	Continuous increase	Increase in flooding and damage. Saline intrusion.	Early warning. Stress tolerant varieties.	More climate-resilient production (i.e. saltwater), and higher and stable income for farmers.

C. Alignment

h. Is the proposed project aligned with country's priorities/ strategies? What current climate-related coping strategies/ interventions are in place and how will the proposed GCF project complement and/or address major gaps?

Proposals should confirm alignment of the proposed activity with the host country's national plans and climate strategies (including their NAPs, NAPAs, long-term climate strategies, and adaptation communications including those submitted as components of NDCs, as applicable) as well as previous/ongoing initiatives/projects and future projects in pipeline (GCF or otherwise). This helps ensure the country ownership of the proposal and that activities are targeting areas of highest potential need and impact for that country.

D. Monitoring and evaluation

i. How will adaptation progress be measured? What are the indicators to be monitored?

Proposals should include a description of the monitoring and evaluation system that will be used to assess the climate impact[9] of the proposed activity and quantify the adaptation beneficiaries. This will facilitate the assessment during implementation of whether the ToC is working, whether the funded activity generated the climate impact expected and will also inform the design of more impactful future adaptation options.

Core indicators for adaptation projects to be monitored are as follows;

- The number of direct and indirect project beneficiaries[10] (Core indicator 2 *''Direct and indirect beneficiaries reached''*).
- The expected reduction in loss of lives, maintenance of the value of physical assets, livelihoods, and/or reductions in environmental or social losses otherwise arising from the impact of extreme climate-related disasters and climate change in the geographical area of the GCF intervention (in relation to the Core indicator 3 *''Value of physical assets made more resilient to the effects of climate change and/ or more able to reduce GHG emissions''* and 4 *''Hectares of natural resource areas brought under improved low emission and/or climate-resilient management practices''* of Integrated Results Management Framework).

FAO's Resilience Index Measurement and Analysis tool (RIMA) provides ideas for how to measure resilience in the context of climate change. FAO's Self-evaluation and Holistic Assessment of Climate Resilience of Farmers and Pastoralists tool (SHARP) can be used for baseline assessments in close cooperation with local farmers and pastoralists. The FAO Adaptation Indicators provide guidance on how

[9] Impact here refers to adaptation beneficiaries as described in the GCF Results Management Framework (RMF), rather than the IPCC. The GCF Integrated Results Management Framework Handbook provides useful information.
[10] Please refer to FAO-OCB's *guidance note on beneficiaries' calculation for FAO-GCF projects*.

adaptation progress can be measured in the context of your project. This section can be very short at CN stage.

Figure 4. Short summary of demonstrating adaptation impact potential[11]

When claiming adaptation impact, the proposals need to clarify adaptation…	**Principles**
• … of **whom/what?** (people, systems) • … **to what?** (current and future climate risks and impacts)	**A: Identification**
• … **when and how?** (this activity/project will reduce climate risks and increase climate resilience)	**B: Response** **C: Alignment**
• … tracking progress **when and how?**	**D: Monitoring and evaluation**

Note that the GCF financing should be tailored to the incremental cost of adaptation to climate issues. Problems that are driven mainly by non-climate drivers should be co-financed.

3.2 Mitigation

a. *What is the emissions profile? Describe country's overall emissions (net or relative to the production, i.e. its overall emission intensity) and contextualize sector(s) emission profile(s) (intensity) that will be targeted by the project. The country (economy-wide) and sector profile serve as baseline to understand the emission reduction and, where applicable, sequestration potential of the project (to be detailed under b.).*

Relevant information to start with can often be found in a country's NAMA, NDC, National Communications to the UNFCCC, and national policies and strategies.

b. *What is the regulatory (legal and policy) framework in place to reduce emissions (and, where applicable increase the sequestration) potential at national and sub-national level?*

Please indicate how the project fits in with the country's national priorities and its full ownership of the concept. Is the project/programme directly contributing to the country's INDC/NDC or national climate strategies or other plans such as NAMAs or equivalent? If so, please describe which priorities identified in these documents the proposed project is aiming to address and/or improve.

c. *What are the mitigation needs? This section needs to be based on the information provided under a. and b.*

d. *What are the barriers to successfully implementing mitigation activities in the context of this project?*

This information is often not fully available at CN stage. However, the CN should include an initial list of potential barriers to successful implementation and how to mitigate any risks to the projects' success related to any of such barriers.

e. *What activities would be implemented under this project to reduce emissions and/ or sequester carbon?*

This section needs to also outline how these activities correspond and/ or help to further advance the regulatory framework in place. In addition, this section needs to clarify how proposed activities would overcome barriers outlined under d. Detailed information on e. will need to be given under B.2.

f. *What is the emission reduction (and where applicable, sequestration) potential of the project activities? In other words, how much emissions can be reduced (or carbon sequestered) due to the activities put*

[11] One of key messages from iTAP during the presentation in July 2023.

in place relative to scenario outlined in a.? Make sure to provide information on how the project contributes to reducing the emission intensity of the country and/ or sector.

The proposal should indicate the potential mitigation impact measured by the estimated quantity of greenhouse gas (GHG) emissions in metric tonnes of carbon dioxide equivalent avoided, reduced, or sequestered (Core Indicator 1) through GCF-funded interventions as compared to a baseline level of GHG emissions. An estimated number of the mitigation impact is required at CN stage.

The baseline scenario is a hypothetical situation for the GCF funded activity that reasonably represents the anthropogenic emissions by sources of GHGs that would occur in the absence of the GCF funded activity. Note this should be a forward-looking counterfactual baseline scenario over a certain time period rather than a single baseline year scenario[12].

It is also important to highlight catalytic effect and scalability, e.g. highlight how the enabling environment (e.g. through policy or access to services) will ensure continuity and further adoption in other areas of the country.

The FAO has a number of concrete tools and project teams that allow for a comparison of between baseline and project scenario conditions: GLEAM-I (for the livestock sector) and EX-ACT[13] (mostly applicable in the AFOLU context). For further guidance – including contacts of relevant colleagues – on these tools please contact your regional GCF focal point in OCB.

g. *How will the mitigation process be measured? This section can be very brief at CN stage.*

h. *What is the overall situation of the country? Makes sure to dedicate one paragraph to the overall (socio-economic) situation of the country (including sub-national information, where possible).*

3.3 Use of information and data

GCF proposals should make use of the best available information and data, which may come from a variety of sources. As many developing countries have stressed the lack of information and data for specific contexts, the GCF recognizes the significant variation in information and data availability and thus makes no prescription of the use of any specific information or data type. However, the GCF requires project-specific information and observational data which may only become available when conducting the feasibility study (which is mandatory for the full Funding Proposal). Where such information and data are not available or are not of sufficient quality then the reason for not using/accessing original/raw/observational datasets should be clearly stated, and alternative peer-reviewed and scientifically credible datasets such as global gridded data or climate reanalyses data may be used to show historical climate trends (the information platforms listed in Annex 1 can be used).

When defining the climate hazard, exposure, and vulnerability from the combination of historical data and future climate projections, it is important to consider the consistency and agreement between the different sources of information including traditional, local and indigenous knowledge and practices. Wherever possible use the same consistent datasets (past and future projections) and analytical approaches throughout the document, otherwise it will be difficult to interpret the results in a coherent way to justify intervention packages and the whole project design.

[12] The GCF Integrated Results Management Framework Handbook

[13] Note that it is important to get clearance from EX-ACT team on calculation by using EX-ACT. Please contact your regional GCF focal point in OCB.

Annex 1 List of tools/information sources (not exhaustive)

Guiding questions	Name	Thematic area	Description/remark	Scale	Type of tool/sources
Adaptation					
(a)	Climate Information Platform (CIP)	• Climate change trend and projection	The CIP co-developed by the GCF and WMO provides a rapid regional assessment of selected climate indicators.	• National • Site-specific	Platform
(a)	World Bank Climate Change Knowledge platform (CCKP)	• Climate change trend and projection	The CCKP contains climate, disaster risk, and socio-economic datasets, as well as synthesis products, such as the Climate Risk Country Profiles.	• National • Sub-national	Platform
(a)	IPCC Assessment Reports (6^{th} Assessment Report, WGI in particular) Interactive IPCC Atlas	• Climate change trend and projection	This report and interactive atlas can provide some useful information on observed and projected climate change.	• Global • Regional • National	Report Platform
(e)	Climate Risk Toolbox (CRTB)	• Climate risk	The CRTB is hosted by the FAO Hand-in-Hand Geospatial Platform and provide quick climate risk assessment.	• National • Sub-national (not all data)	Platform
(a)	FAO CAVA Platform FAO CAVAnalytics	• Climate change trend and projection	CAVA platform provides the most essential climate change information in a user-friendly interface. The more advance CAVA Analytics tool can also be provided upon request (contact OCB Risk Team).	• National • Sub-national	Platform
(b) (c)	Emergency Events Database (EM-DAT)	• Climate induced disasters	EM-DAT launched by the Centre for Research on the Epidemiology of Disasters (CRED) contains essential core data on the occurrence and effects of over 22,000 mass disasters in the world from 1900 to the present day.	• National • Sub-national	Database
(b) (c)	DesInventar[14]	• Climate induced disasters	DesInventor provides data about damage, losses and the effects of disasters.	• National • Sub-national	Database
(b) (c)	WMO	• Climate induced disasters	WMO's report provides an overview of mortality and economic losses from weather, climate and water related hazards from 1970 to 2019.	• Global • Regional	Report

[14] A new disaster losses and damages tracking system will replace the DesInventar once it becomes available.

Guiding questions	Name	Thematic area	Description/remark	Scale	Type of tool/sources
(b)	Global Agro-Ecological Zones (GAEZ) assessment	• Historical and projected impacts of climate change	The GAEZ v4 spatial data are organized in six themes: (1) Land and Water Resources, (2) Agro-climatic Resources, (3) Agro-climatic Potential Yield, (4) Suitability and Attainable Yield, (5) Actual Yields and Production, and (6) Yield and Production Gaps	• National • Sub-national	Platform
(b)	Climate Adaptation in Rural Development (CARD) Assessment Tool	• Projected impacts of climate change	This tool developed by IFAD can provide information on the effects of climate change on the yield of major crops.	• National	Excel-based information
(b) (f)	FAO's report "Managing risks to build climate-smart and resilient agrifood value chains" (2022)	• Impacts of weather hazards	This report can give you an idea of possible climate- and weather-related hazards and impacts along the agrifood value chain, besides recognizing the added value of climate services on building climate resilience	• Global	Report
(b) (f)	Global Outlook on Climate Services in Agriculture: Investment opportunities to bridge the last-mile gap (2021).	• Impacts of weather hazards	This report provides an outlook of the state of weather-informed agricultural advisories worldwide, including regional barriers and opportunities along the climate services value chain	• Global • Regional • National	Report
(d) (i)	FAO's Resilience Index Measurement and Analysis tool (RIMA)	• Measurement of adaptation progress	RIMA provides ideas for how to measure resilience in the context of climate change	• Household level	Methodology
(d) (i)	FAO's Self-evaluation and Holistic Assessment of Climate Resilience of Farmers and Pastoralists tool (SHARP)	• Measurement of adaptation progress	SHARP tool is a holistic assessment of farmers climate resilience at the household level. The improved tool (SHARP+) is available.	• Household level	Methodology
Various	AQUASTAT	• Statistics		• National	Database
Various	FAOSTAT	• Statistics		• National	Database
Mitigation					
(f)	EX-ACT	• GHG emission calculation	EX-ACT is the GHG accounting tool that covers the entire agricultural sector.	• Project specific	Methodology
(f)	GLEAM-I	• GHG emission calculation	GLEAM is a modelling framework that simulates the interaction of activities and processes involved in livestock production and the environment.	• National • Sub-national	Methodology
(f)	FAOSTAT	• Statistics		• National	Database

Annex 2 High-level principles for demonstrating the climate impact potentials of GCF supported activities

The following high-level principles have been set out in GCF/B.33/05 to guide the overarching narrative of a proposal which ensures that it addresses climate change and thus contributes to the criteria of the GCF Investment Framework.

Adaptation

For adaptation activities, climate impact potential is established by providing an analysis to show that a proposed activity is likely to be an effective adaptive response to the risk or impact of a specific climate change hazard. Establishing the impact potential for an adaptation funding proposal follows four high-level principles:

1. **Identification**: Adaptation proposals should show how the activity addresses current or future projected climate change risk or impact, and why it is likely to be an effective response. Proposals should identify the systems at risk and the climate change hazard affecting them or expected to in the future. They should show how climate change has contributed, or will contribute, to the specific risk or impact that the proposed activity addresses using the best available information. Where relevant, proposals should also consider any non-climatic factors that may be causing or exacerbating the risk or impact and describe the interactions between climate change and non-climatic drivers. Vulnerability assessments can be used to identify groups, sectors and subregions most susceptible to the climate change impact and therefore will provide information to select and prioritise appropriate adaptation outcomes.

2. **Response**: Proposals should explain how the activity will reduce the exposure and/ or vulnerability (of people, systems, or ecosystems) and thus lessen the climate change risk or impact. Where relevant, a justification should be given for why the proposed activity was selected over alternatives. Proposals should consider barriers (e.g., technical, social, institutional, regulatory) to the implementation of the activity and describe how the project aims to overcome those barriers. Proposals should apply a methodological approach for the quantification of the number of beneficiaries expected to result from the activity.

3. **Alignment**: Proposals should confirm alignment of the proposed activity with the host country's national plans and climate strategies (including their NAPs, NAPAs, long-term climate strategies, and adaptation communications including those submitted as components of NDCs, as applicable). This helps ensure the country ownership of the proposal and that activities are targeting areas of highest potential need and impact for that country.

4. **Monitoring and evaluation**: Projects with a well-designed theory of change are more likely to result in successful outcomes that can be measured and evaluated. Proposals should include a description of the monitoring and evaluation system that will be used to assess the outcomes of adaptation activities and to quantify the adaptation beneficiaries. This will facilitate the assessment during implementation of whether the funded activity generated the climate impact expected and will also inform the design of more impactful future adaptation options.

Mitigation

Proposals for mitigation activities need to demonstrate that a projected level of GHG emissions reductions (or removals) will occur, and that these emission reductions would not have happened without the GCF-funded activity. Establishing the climate impact potential for a mitigation funding proposal involves the following high-level principles which mirror established procedures and best-practices in emission reductions estimation across the climate finance landscape:

1. Proposals should confirm **alignment of the activities with host country priorities**, including its nationally determined contribution (NDC) or other national and long-term climate strategies consistent with the long-term global goal to hold the increase in the global average temperature to well below 2°C

and to pursue efforts to limit the temperature increase to 1.5°C. The GCF is an operating entity of the financial mechanism of the UNFCCC and the Paris Agreement, and it is important to confirm that the GCF-funded activity is aligned with the NDC or other national climate strategies of the country. This also helps ensure that country ownership is integrated in the proposal and that activities are targeting the areas of highest potential impact and need for that country;

2. **A methodological approach for the quantification of the mitigation results of the activity and its monitoring** needs to be selected and implemented. The GCF does not prescribe any specific methodologies, but strongly encourages AEs to utilize, whenever possible, the multitude of tools and methodologies developed for the quantification and monitoring of mitigation impact. Since the adoption of the UNFCCC and subsequently the Kyoto Protocol and the Paris Agreement, significant work has been done towards establishing methodologies for mitigation activities. Examples of suitable methods include the Clean Development Mechanism, and Joint Implementation under the Kyoto Protocol, which have established methodologies for quantifying mitigation impact for projects and programmes.

3. Proposals should **use the methodology most relevant to the specific activities proposed**. Articulation and assessment of mitigation impact follows a number of standard steps: determine project impact boundaries; define the baseline; and show additionality. In the context of a mitigation project, an activity is considered additional if it can be shown that the GHG emission reductions would not occur in the absence of the GCF funding. Each of these steps is described in more detail and illustrated with worked examples in supplementary online resources.

4. The quantification of mitigation impact should **use consistent assumptions (e.g., emission factors) to those made in national GHG reporting** as this will allow for the accurate quantification of the support provided to countries in meeting their NDCs in line with the Paris Agreement.

5. Proposals should describe **the establishment of a Measurement, Reporting, and Verification (MRV) system** for the GHG emission reductions and removals of the proposed activity. This will facilitate the assessment, during and after implementation, of whether the funded activity generated the projected mitigation results. This will in turn provide learnings to the GCF, AEs, and host countries towards maximising the impact of future mitigation activities. When describing the MRV system for mitigation results, proposals should include all indicators, equations, input values to formulae, and any other assumptions used to quantify the emission reductions or removals, baseline and project scenarios, and information on how the MRV will be conducted. Proposals should also provide projections of the annual emission reductions or removals during the lifetime of the project or programme.